

PROBLEM STATEMENT

The N-Queens Problem is a classical Artificial Intelligence problem in which **N queens** must be placed on an **$N \times N$ chessboard** such that no two queens attack each other. A queen can attack any other queen placed in the same row, same column, or along the diagonals. The objective is to find a valid arrangement of all N queens while satisfying these constraints. The problem is widely used to demonstrate constraint satisfaction and search techniques in Artificial Intelligence.